MEGATHON X 2026 — M2 MID-SUBMISSION

Project: ShopSaarthi – Accessible Voice-First Shopping Assistant TEAM NO.293

1. Current Approach and Changes Since M1

ShopSaarthi is an accessible, voice-first shopping assistant designed to make online shopping easier and more convenient. Since M1, development has moved from initial setup toward connecting the interface, backend, product data, and voice features into a working prototype.

- **Frontend and UI:** Developed product cards, category filters, search, and a shopping cart. Added quantity controls, order confirmation, shopping settings, language selection, larger text, and high-contrast options.
- **Backend and product data:** Developed APIs to retrieve the product list and individual product details. Added sample data for product names, categories, brands, prices, ratings, features, reviews, and stock. Created a product-comparison API with input validation.
- **Voice and languages:** Added a voice-assistant interface with voice controls and speech-speed settings. Language preferences include English, Hindi, and Telugu. Bhashini multilingual voice integration still needs completion and validation.

2. Experiments Run and Results

The following are early results from the demonstrated tests—not evidence that the full system is integrated:

- **Home API:** Returned HTTP 200 with “ShopSaarthi AI backend is running”.
- **Product listing API:** Returned HTTP 200 and displayed six sample products with details.
- **Individual product API:** Successfully retrieved a product by its ID in the tested case.
- **Shopping interface demo:** Showed product browsing, category selection, adding items to the cart, quantity controls, and order confirmation.
- **Accessibility settings:** Displayed English, Hindi, and Telugu choices, plus larger-text and high-contrast options.
- **Comparison API:** Returned HTTP 400 with a validation message when fewer than two products were selected. A valid comparison with at least two products still needs testing.

3. Progress on Testing

- Basic backend API responses and individual product retrieval work in the tested cases; the interface demonstrates core browsing and cart interactions.
- Further testing is needed for frontend–backend integration, database consistency, valid product comparisons, and the complete order flow.
- Multilingual speech recognition and spoken responses need testing across English, Hindi, and Telugu.
- End-to-end reliability and accessibility also remain to be verified.

4. Plan for the Remaining Time

1. Complete and improve applicable AI features, including product search, recommendations, and comparison.
2. Continue Bhashini integration for multilingual voice input and spoken responses.
3. Connect the frontend, backend, database, and voice features into one working application.
4. Test the complete user journey, fix bugs, and improve accessibility and reliability.
5. Prepare the final prototype, clean code, final report, and demonstration video.

Current progress estimate: The team estimates AI features and Bhashini multilingual voice integration are approximately **50–60% complete**. This is a team estimate, not an independently verified result.